

Validation Source Capability & Competence Module (VSCCM)

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: VALIDOS® — Validation Governance Architecture

Parent Standard: Validation Source Reliability Standard (VSRS)

Operational Layer: Validation Source Reliability Governance Layer

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Governed Space: Validation Source Capability & Competence

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· ORA™ · AGA™ · AIG® · CLIA® · MGIA™ · ASGA™ · RIS™

AI-Readable: Yes

Authority: OOF®

Protection: MIP® — Methodological Intellectual Property

Canonical Language: English (UCL™)

Governed Space

Validation Source Capability & Competence

Canonical Definition

Validation Source Capability & Competence Module (VSCCM) defines the governance framework for determining whether an identified Validation Source possesses the technical, operational, methodological, informational, professional, analytical, observational, measurement, or domain-specific capability and competence necessary to perform the evidentiary role assigned to it within a governed validation process.

It establishes the conditions under which an identifiable and traceable source may be recognized as **capable and sufficiently competent for a specific validation-source role**.

Operational Role

VSCCM serves as the third internal governance space of the Validation Source Reliability Standard.

The preceding modules establish:

VSIAM — Who or what is the source, and can the evidence be attributed to it?

VSPLM — Where did the evidence originate, how did it reach validation, and upon which upstream sources does it depend?

VSCCM asks:

Is the source actually capable and sufficiently competent to perform the evidentiary role assigned to it?

The governed progression is:

Identified Source → Defined Source Role → Capability Requirements → Capability Assessment → Competence Requirements → Competence Assessment → Role Fitness Determination

VSCCM does not determine overall Source Reliability.

It establishes whether the source possesses the functional capability and competence required for its assigned role.

Module Operational Space

VSCCM governs:

- validation-source capability,
- source competence,
- role capability,
- technical capability,
- measurement capability,
- observational capability,
- operational capability,
- analytical capability,
- methodological capability,
- informational capability,
- professional competence,
- domain competence,
- system capability,
- instrument capability,
- capability boundaries,
- competence boundaries,
- capability evidence,
- competence evidence,
- capability limitations,
- competence limitations,
- capability uncertainty,
- competence uncertainty,
- capability degradation,
- competence degradation,
- capability change,
- competence change,
- and capability-and-competence traceability.

Module Function

VSCCM applies after source identity, attribution, provenance, and material lineage have been sufficiently established.

Its function is to prevent:

- source reputation being mistaken for functional capability,
- institutional authority being mistaken for technical competence,
- professional qualification being assumed relevant to every domain,
- a measuring instrument being used outside its validated range,
- an AI model being treated as competent outside its demonstrated capabilities,
- access to information being mistaken for ability to interpret it,
- operational experience being mistaken for methodological expertise,
- historical competence being assumed current,
- capability in one role being transferred automatically to another,
- and source confidence being mistaken for source competence.

Capability and Competence Are Related but Distinct

VALIDOS® distinguishes:

Capability

from:

Competence

Capability asks:

Can the source technically, operationally, informationally, or functionally perform the evidentiary task?

Competence asks:

Does the source possess sufficient knowledge, expertise, methodological understanding, qualification, or demonstrated proficiency to perform the role appropriately?

A source may possess one without fully possessing the other.

Capability Is Role-Specific

Capability is not a universal property.

A source may be highly capable in one role and incapable in another.

For example:

A sensor may be highly capable of measuring temperature.

It may have no capability to determine structural integrity.

A financial analyst may be highly competent in market analysis.

That does not establish medical competence.

An AI model may perform strongly in document classification.

That does not establish capability for autonomous safety decisions.

VALIDOS® therefore establishes:

Capability must be assessed against the exact evidentiary role assigned to the source.

Source Role Requirements

Before capability can be assessed, the source role must be sufficiently defined.

Possible roles may include:

- measurement,
- observation,
- recording,
- analysis,
- interpretation,
- classification,
- aggregation,
- transformation,
- expert judgment,
- monitoring,
- data provision,
- operational reporting,
- or another evidentiary function.

Each role may create different capability and competence requirements.

Technical Capability

Technical capability concerns whether a source possesses the technical means necessary to perform its assigned role.

Relevant considerations may include:

- hardware capability,
- software capability,
- system functionality,
- computational capability,
- instrument specification,
- measurement range,
- resolution,
- detection limits,

- interoperability,
- or technical operating conditions.

Technical capability should be assessed relative to the validation requirement rather than assumed from system availability.

Measurement Capability

Where a source generates measurements, VSCCM examines whether it is capable of measuring the relevant property with sufficient:

- range,
- sensitivity,
- precision,
- resolution,
- repeatability,
- stability,
- and operational suitability.

The Evidence Adequacy Standard governs whether the resulting evidence is adequate.

VSCCM governs whether the source itself is capable of producing the type of measurement required.

Observational Capability

Some sources generate evidence through observation rather than direct measurement.

Observational capability may depend upon:

- access,
- visibility,
- detection ability,
- observation conditions,
- temporal presence,
- system coverage,
- or another factor affecting what the source could actually observe.

A source cannot reliably report conditions it had no meaningful capability to observe.

Operational Capability

A source may need sufficient operational access or operational functionality.

For example:

A monitoring system may be technically sophisticated but unable to observe a subsystem outside its telemetry coverage.

An employee may understand an operation but lack access to the relevant event.

An external auditor may possess expertise but not direct operational visibility.

VSCCM distinguishes theoretical capability from actual operational capability.

Informational Capability

A source may possess competence but lack access to the information necessary to perform the assigned role.

Informational capability asks whether the source had sufficient access to:

- relevant data,
- records,
- system states,
- documentation,
- events,
- or other required information.

This prevents expertise from compensating for absence of information.

Analytical Capability

Where a source performs analysis, VSCCM evaluates whether it possesses sufficient capability to execute the required analytical function.

This may concern:

- statistical analysis,
- computational analysis,
- causal analysis,
- comparative analysis,
- risk analysis,
- pattern recognition,
- model evaluation,
- or another defined analytical task.

The correctness of the Validation Method remains governed separately.

VSCCM determines whether the source can legitimately perform the

analytical role assigned to it.

Methodological Capability

A source may possess domain knowledge while lacking the methodological capability required by a specific validation process.

Methodological capability may concern:

- experimental design,
- statistical reasoning,
- validation methodology,
- evidence interpretation,
- uncertainty analysis,
- sampling,
- simulation,
- or another relevant methodological function.

VSCCM therefore prevents domain expertise from automatically implying methodological competence.

Professional Competence

Where professional judgment materially contributes to Validation Evidence, VSCCM may assess:

- qualifications,
- certifications,
- professional standing,
- relevant experience,
- demonstrated proficiency,
- specialized training,
- or equivalent competence indicators.

Formal credentials may support competence.

They do not automatically establish competence for every assigned role.

Domain Competence

Domain competence concerns whether the source possesses sufficient understanding of the subject matter relevant to the evidentiary role.

Domains may include:

- medicine,
 - finance,
 - engineering,
-

- cybersecurity,
- law,
- industrial operations,
- artificial intelligence,
- infrastructure,
- or another specialized field.

Competence should remain bounded to the domain actually demonstrated.

Experience and Competence

Experience can strengthen competence.

However:

Experience ≠ Automatic Competence

Long experience with an outdated methodology may not establish current competence.

Likewise, limited formal experience may coexist with strong demonstrated capability.

VSCCM therefore allows multiple forms of competence evidence rather than relying upon one indicator.

Certification and Competence

Certification may provide evidence of competence within its defined scope.

VALIDOS® establishes:

Certification ≠ Universal Competence

A certification may be:

- scope-limited,
- time-limited,
- jurisdiction-limited,
- version-specific,
- role-specific,
- or dependent upon continued conditions.

VSCCM therefore examines what the certification actually demonstrates.

Authority and Competence

Authority may permit a source to act.

Competence determines whether the source can perform the assigned evidentiary role appropriately.

Therefore:

Authority to Act ≠ Competence to Validate

An institution may possess legal authority while relying upon specialized experts or technical systems for evidentiary functions.

VSCCM preserves this distinction.

AI Source Capability

Where an AI model or AI system acts as a Validation Source, capability assessment may include:

- model function,
- model version,
- demonstrated task performance,
- context-window limitations,
- modality limitations,
- tool access,
- information access,
- known failure modes,
- uncertainty behavior,
- and operating constraints.

Model scale, popularity, or general benchmark performance does not automatically establish capability for the specific validation role.

Autonomous Source Capability

Where an autonomous system generates Validation Evidence, capability assessment may additionally consider:

- sensor access,
- decision boundaries,
- operating permissions,
- environmental awareness,
- state visibility,
- system dependencies,
- and operational constraints.

Autonomy does not establish evidentiary competence by itself.

Instrument Capability

Measurement instruments may possess defined operating boundaries.

These may include:

- calibration range,
- environmental tolerance,
- sensitivity,
- measurement uncertainty,
- operating temperature,
- sampling frequency,
- or another technical constraint.

Evidence generated outside those boundaries may require qualification.

Capability Boundary

A **Capability Boundary** defines the conditions within which a source can legitimately perform its evidentiary role.

The boundary may concern:

- task type,
- domain,
- measurement range,
- environment,
- population,
- system version,
- information access,
- operating state,
- or another relevant condition.

Capability must not silently extend beyond demonstrated boundaries.

Competence Boundary

A **Competence Boundary** defines the scope within which demonstrated knowledge, expertise, qualification, or proficiency remains relevant.

For example:

A specialist may possess deep competence within one medical discipline but not another.

An AI model may demonstrate competence on one task family but not on a materially different task.

Competence should therefore remain explicitly bounded.

Capability Evidence

Capability claims should themselves possess supporting evidence where material.

Capability evidence may include:

- technical specifications,
- calibration records,
- performance tests,
- proficiency results,
- operational history,
- benchmark results,
- certification,
- controlled demonstrations,
- or other appropriate evidence.

VSCCM does not require one universal evidence type.

The evidence should correspond to the capability claim being made.

Competence Evidence

Competence evidence may include:

- qualifications,
- certifications,
- training records,
- demonstrated performance,
- relevant experience,
- professional assessments,
- peer review,
- proficiency testing,
- or another defensible competence indicator.

The strength required should remain proportionate to the validation role.

Self-Declared Capability

A source may claim its own capability or competence.

Self-declaration may provide information.

It does not automatically establish the claim.

Where validation consequence is significant, independent or externally verifiable capability evidence may be required.

Capability Limitation

A source may be capable overall while possessing known limitations.

Examples include:

- reduced precision at boundary conditions,
- limited language coverage,
- limited population representation,
- inability to observe certain system states,
- limited historical access,
- or reduced performance under degraded conditions.

Limitations must remain visible in subsequent Source Reliability governance.

Competence Limitation

Competence may also be limited.

Examples include:

- narrow specialization,
- outdated training,
- insufficient experience with a new system,
- lack of methodological expertise,
- or incomplete understanding of a relevant domain.

A limitation does not automatically make the source unusable.

It constrains the role for which competence can legitimately be claimed.

Capability Uncertainty

Where available information is insufficient to establish capability, VSCCM preserves:

Capability Undetermined

rather than assuming capability from reputation, authority, or availability.

Competence Uncertainty

Where competence cannot be sufficiently established, VSCCM preserves:

Competence Undetermined

This may require:

- additional evidence,
- independent assessment,
- restricted source role,
- corroboration,
- or another governance response.

Capability Status

VSCCM may establish:

Capability Established

The source possesses sufficient capability for the defined evidentiary role.

Capability Conditionally Established

Capability exists only within specified boundaries or conditions.

Capability Undetermined Available information is insufficient to determine capability.

Capability Insufficient

The source does not possess sufficient capability for the proposed role.

Competence Status

VSCCM may establish:

Competence Established

The source possesses sufficient competence for the assigned evidentiary role.

Competence Conditionally Established

Competence is sufficient only within defined conditions or boundaries.

Competence Undetermined

Available information is insufficient to establish competence.

Competence Insufficient

The source does not possess sufficient competence for the assigned role.

Role Fitness

Capability and competence are integrated into a **Source Role Fitness** assessment.

Possible outcomes may include:

Role Fit

Conditionally Role Fit

Role Fitness Undetermined

Role Unfit

Role Fitness remains an input into overall Source Reliability Determination.

It is not itself a Source Reliability status.

Capability Degradation

Capability may degrade because of:

- equipment aging,
- calibration loss,
- software degradation,
- infrastructure failure,
- reduced information access,
- model drift,
- system modification,
- environmental change,
- or another material condition.

Historical capability should not automatically be treated as permanent.

Competence Degradation

Competence may degrade because of:

- outdated knowledge,
- expired qualifications,
- loss of practice,
- domain evolution,
- methodology change,
- or changed role requirements.

Competence may therefore require periodic reassessment.

Capability Change

Capability may improve or decline because of:

- system upgrade,
- model update,
- new tools,
- additional access,
- new sensors,
- improved infrastructure,
- changed configuration,
- or altered operating conditions.

Material changes must remain traceable.

Competence Change

Competence may change through:

- training,
- certification,
- experience,
- role change,
- specialization,
- methodological development,
- or loss of required qualification.

VSCEM preserves the temporal relationship between competence and the evidence produced.

Capability at Time of Evidence Generation

A critical rule is:

Current capability does not automatically prove historical capability.

The relevant question may be whether the source possessed sufficient capability **when the evidence was generated**.

Likewise:

Historical capability does not automatically establish current capability.

VSCEM therefore keeps capability temporally situated.

Competence at Time of Contribution

Where competence materially affects evidence, the relevant competence state is the one existing when the source performed the evidentiary role.

Subsequent qualification cannot automatically repair an earlier competence deficiency.

Capability and Evidence Adequacy

VSCCM maintains a strict boundary with VEFS.

VEFS asks:

Is the resulting evidence adequate?

VSCCM asks:

Was the source capable and competent to perform the role that produced or handled that evidence?

A capable source may produce inadequate evidence.

An apparently adequate evidence item may have been produced by a source whose capability cannot be established.

Both conditions require separate governance.

Capability and Source Reliability

Capability is necessary for many source roles.

But:

Capability ≠ Source Reliability

A technically capable source may still be:

- conflicted,
- dependent,
- manipulated,
- compromised,
- or otherwise unreliable.

VSCCM therefore provides one component of Source Reliability governance.

Minimum Implementation Framework

1. Establish Source Role

Identify the exact evidentiary function performed by the source.

2. Establish Capability Requirements

Determine the technical, operational, informational, analytical, methodological, or other capabilities required for that role.

3. Assess Source Capability

Evaluate whether the source possesses the required capability within the relevant conditions.

4. Establish Competence Requirements Where judgment or expertise matters, identify the knowledge, qualification, experience, or proficiency required.

5. Assess Source Competence

Evaluate the available competence evidence.

6. Establish Boundaries and Limitations

Define material:

- capability boundaries,
- competence boundaries,
- limitations,
- conditions,
- and uncertainty.

7. Establish Capability & Competence Status

Classify the relevant source conditions.

8. Determine Source Role Fitness

Determine whether the source is:

- Role Fit,
- Conditionally Role Fit,
- Role Fitness Undetermined,
- or Role Unfit.

9. Preserve Lifecycle Traceability

Maintain:

- Validation Source reference,
- Source Role,
- capability requirements,
- capability evidence,
- Capability Status,
- competence requirements,

- competence evidence,
- Competence Status,
- capability boundaries,
- competence boundaries,
- limitations,
- uncertainty,
- Role Fitness,
- evidence-generation time,
- change history,
- reassessment triggers,
- and sufficient lifecycle traceability.

Capability & Competence Outputs

VSCCM may produce:

- Source Capability Profile,
- Source Capability Assessment,
- Source Competence Profile,
- Source Competence Assessment,
- Technical Capability Record,
- Measurement Capability Record,
- Observational Capability Record,
- Operational Capability Record,
- Informational Capability Record,
- Analytical Capability Record,
- Methodological Capability Record,
- Capability Boundary,
- Competence Boundary,
- Capability Limitation Record,
- Competence Limitation Record,
- Capability Status,
- Competence Status,
- Source Role Fitness Determination,
- Capability Change Record,
- Competence Change Record,
- and Capability & Competence Reassessment Trigger.

Use Case 1 — Industrial Sensor

Scenario

A validation relies upon measurements from an industrial sensor.

Application

VSCCM determines that the sensor is calibrated and technically capable within a defined measurement range.

Some measurements were generated beyond that range.

Result

The sensor may remain Role Fit within its established Capability Boundary while measurements outside that boundary cannot derive source-capability confidence from the same determination.

Use Case 2 — AI Validation Source

Scenario

An AI model is used to classify technical incident reports during validation.

Application

VSCCM evaluates the specific model version, demonstrated classification performance, domain limitations, language coverage, information access, and known failure conditions.

Result

Strong general model performance is not treated as universal competence.

The AI source receives a role-specific capability determination.

Use Case 3 — Expert Assessment

Scenario

A highly experienced engineer provides an expert interpretation of a medical-device software failure.

Application

VSCCM distinguishes engineering competence from the specialized medical and clinical competence potentially required for parts of the assessment.

Result

The expert may be Role Fit for technical failure analysis while additional expertise is required for clinical conclusions.

Architectural Position

Validation Source Capability & Competence is the third internal governance space of the **Validation Source Reliability Standard (VSRS)**.

The progression now becomes:

Source Identity & Attribution → Source Provenance & Lineage → Source Capability & Competence

VSIAM establishes:

Who or what is the source?

VSPLM establishes:

Where did its evidence originate and upon what does it depend?

VSCCM establishes:

Can the source actually perform the evidentiary role assigned to it?

The next governance stage can then examine:

Is the source sufficiently independent, and what conflicts, incentives, or dependencies may affect its evidentiary role?

The VSRS progression therefore continues:

Identity & Attribution → Provenance & Lineage → Capability & Competence → Independence & Conflict → Source Reliability Determination

Foundational Principle

A source should not be trusted to perform an evidentiary role merely because it exists, is authoritative, or appears credible; it must be capable and sufficiently competent for that role.

Canonical Closing Statement

Validation Source Capability & Competence Module establishes the functional qualification layer of source reliability governance within VALIDOS® by determining whether Validation Sources possess the technical, operational, informational, analytical, methodological, professional, and domain-specific capability and competence required for their assigned evidentiary roles. By governing role requirements, demonstrated capability, competence evidence, boundaries, limitations, uncertainty, degradation, change, temporal fitness, and Source Role Fitness, VSCCM prevents reputation, authority, access, certification, or confidence from substituting for demonstrated ability to perform the evidentiary function upon which validation depends.

Parent Resources

[→ VALIDOS® Validation Governance Architecture — Validation Governance Layer](#)

[→ VALIDOS® Architecture Map](#)

[→ VALIDOS® Complete Standards & Modules Index](#)

Internal Module Flow

[→ Previous module: Validation Source Provenance & Lineage Module \(VSPLM\)](#)

[→ Next module: Validation Source Independence & Conflict Module \(VSICM\)](#)

Related Documents

[→ Validation Source Reliability Standard](#)

[→ About Validation Source Reliability Standard](#)

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Canonical Source

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